



Q. I am trying to build a project with MATLAB that uses colour detection at its core. I have worked on various algorithms available on the Internet but they do not seem to work for me. Can you help me with detailed steps for colour detection using MATLAB?

Pooja

Through e-mail

A. Colour detection through MATLAB is fairly easy with all the required functions already inbuilt. The basic algorithm is as follows:

1. Primary colours such as red, green and blue can be detected easily. The algorithm actually works in separate steps.

2. First take a snapshot from a real-time video. The function that you can use for it is mentioned below. This will

immediately return one single image frame from the video input object as shown in Fig. 1.

3. Convert this RGB snapshot to gray. The function used for this process is mentioned below. The output is shown in Fig. 2.

4. Extract the red/green/blue colour components from the original RGB snapshot.

Function `data(:,1)` extracts all the red colour components from the RGB image. Similarly, for green use `data(:,2)` and for blue `data(:,3)`. The output is shown in Fig. 3.

5. Subtract the red-colour components from the gray-colour snapshot using below-mentioned function. You will get the output image as shown in Fig. 4.

6. Now remove the noise from the image by using filter function shown below. Median filtering is a nonlinear operation which is often used in image processing to reduce 'salt and pepper' noise. A median filter is more effective than convolution when the goal is to simultaneously reduce noise and preserve edges.

7. Convert the filtered image into binary image using the binary function shown below. You will get a bright image at the place of red object. The output image `diff_m` replaces all pixels in the input image with luminance greater than level with the value 1 (white) and replaces all other pixels with the value 0 (black) as shown in Fig. 5.



Fig. 1: Snapshot from a real-time video

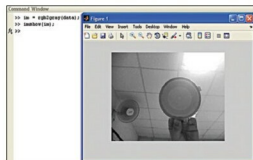


Fig. 2: Gray image

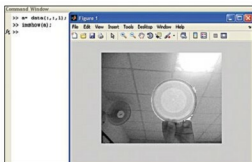


Fig. 3: RED component extracted from RGB image

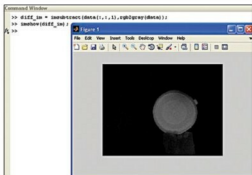


Fig. 4: Subtracting red from gray image

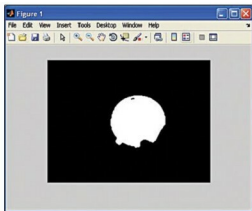


Fig. 5: Converting gray image to binary image

```
diff_m = im2bw(diff_m, 0.18)
```

8. Now you can measure the parameters of that bright image and place a rectangular box over it using mathematical calculations which will show the presence of red colour at that place.

```
stats = regionprops(diff_m, 'BoundingBox','Centroid')
```

That's it!

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender